

2026

■ Pleasanton Math Tournament ■  
Target Round  
Problems 1 & 2

|  |   |
|--|---|
|  | 0 |
|  | 1 |
|  | 2 |
|  | 3 |
|  | 4 |
|  | 5 |
|  | 6 |
|  | 7 |
|  | 8 |
|  | 9 |

Name \_\_\_\_\_

School \_\_\_\_\_

**DO NOT BEGIN UNTIL YOU ARE INSTRUCTED TO DO SO.**

This section of the competition consists of eight problems, which will be presented in pairs. Work on one pair of problems will be completed and answers will be collected before the next pair is distributed. The time limit for each pair of problems is six minutes. The first pair of problems is on the other side of this sheet. When told to do so, turn the page over and begin working. This round assumes the use of calculators, and calculations also may be done on scratch paper, but no other aids are allowed. All answers must be complete, legible and simplified to lowest terms. Record only final answers in the blanks in the left-hand column of the problem sheets. If you complete the problems before time is called, use the time remaining to check your answers.

| Problem 1 | Problem 2 | Scorer's Initials |
|-----------|-----------|-------------------|
|           |           |                   |
|           |           |                   |



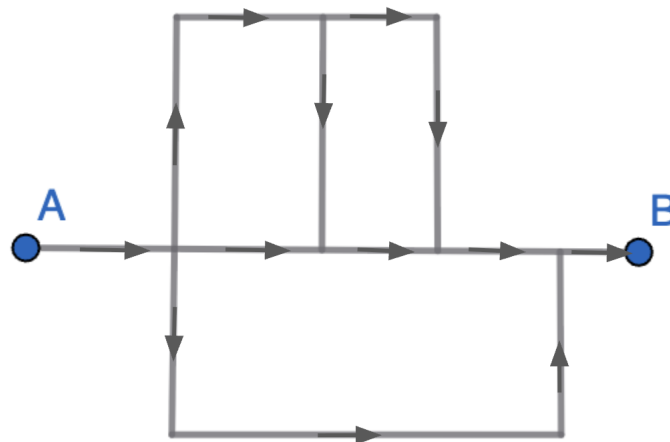
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1. \_\_\_\_\_ pints

Given that there are 4 quarts in 1 gallon and that there are 2 pints in 1 quart, how many pints are in 3 gallons?

2. \_\_\_\_\_

How many paths exist from point A to point B traveling along the roads shown? Each road may only be traversed in one direction, as indicated by the arrows on each road.





## Pleasanton Math Tournament

2026

### ■ Pleasanton Math Tournament ■ Target Round Problems 3 & 4

Name \_\_\_\_\_

School \_\_\_\_\_

**DO NOT BEGIN UNTIL YOU ARE INSTRUCTED TO DO SO.**

|  |   |
|--|---|
|  | 0 |
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|  | 2 |
|  | 3 |
|  | 4 |
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|  | 6 |
|  | 7 |
|  | 8 |
|  | 9 |

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| Problem 3 | Problem 4 | Scorer's Initials |
|-----------|-----------|-------------------|
|           |           |                   |
|           |           |                   |



**Jane  
Street**



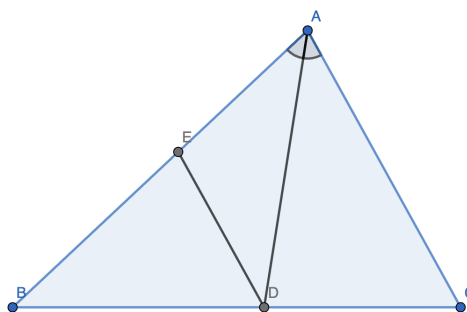
**WOLFRAM**

3. \_\_\_\_\_

What is the least value of  $x$  that satisfies  $x^2y^2 = 2025$  if  $y = 15$ ?

4. \_\_\_\_\_

In  $\triangle ABC$ ,  $\overline{AD}$  bisects  $\angle A$  and meets  $\overline{BC}$  at  $D$ . A point  $E$  lies on  $\overline{AB}$  such that  $\overline{DE} \parallel \overline{AC}$ . Given  $AB = 18$ ,  $AC = 12$ , and  $BC = 20$ , find the length of  $BE$ . Note that the below diagram is not drawn to scale.





## Pleasanton Math Tournament

2026

■ Pleasanton Math Tournament ■  
Target Round  
Problems 5 & 6

Name \_\_\_\_\_

School \_\_\_\_\_

**DO NOT BEGIN UNTIL YOU ARE INSTRUCTED TO DO SO.**

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|  | 0 |
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| Problem 5 | Problem 6 | Scorer's Initials |
|-----------|-----------|-------------------|
|           |           |                   |
|           |           |                   |



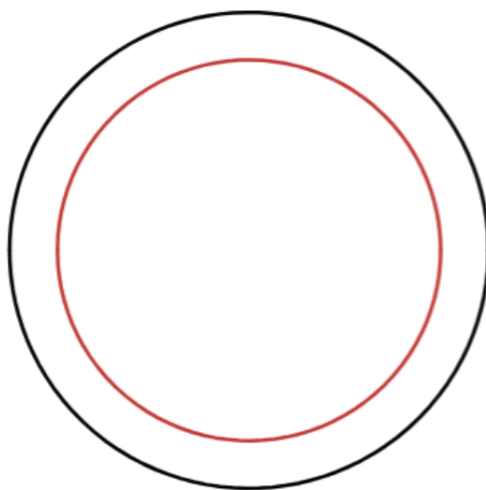
Jane  
Street



WOLFRAM

5. \_\_\_\_\_ units

Two concentric circles with radius 25 and 26 are drawn. To the nearest hundredth, what is the length of the largest line segment that can be drawn in the region strictly between the two circles?



6. \_\_\_\_\_

What is the remainder when the sum of the first 6741 positive cubes is divided by 6?



## Pleasanton Math Tournament

2026

■ Pleasanton Math Tournament ■  
Target Round  
Problems 7 & 8

Name \_\_\_\_\_

School \_\_\_\_\_

**DO NOT BEGIN UNTIL YOU ARE INSTRUCTED TO DO SO.**

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|  | 9 |

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| Problem 7 | Problem 8 | Scorer's Initials |
|-----------|-----------|-------------------|
|           |           |                   |
|           |           |                   |



Jane  
Street



WOLFRAM

7. \_\_\_\_\_

A bag contains five marbles numbered 1 through 5. Three marbles are drawn without replacement and arranged in the order drawn to form a three-digit number. What is the probability that the resulting three-digit number is divisible by 4?

8. \_\_\_\_\_

Shreyas creates a function  $p(n)$  which calculates the square root of the product of the positive integer divisors of  $n$ . Define a new function  $q(n)$  by

$$q(n) = \begin{cases} p(n), & \text{if } p(n) \text{ is an integer,} \\ 1, & \text{otherwise.} \end{cases}$$

For how many positive integers less than or equal to 50 is  $q^{2026}(n) = n$ ? Here  $q^{2026}(n)$  denotes the result of applying the function  $q$  to  $n$  a total of 2026 times. To be precise,  $q^{2026}(n) = \underbrace{q(q(q(\cdots q(n) \cdots)))}_{2026 \text{ times}}$ .